



Empirical approximations for Hoeffding's test of bivariate independence using two Weibull extensions

Gregory E. Wilding^{a,*}, Govind S. Mudholkar^b

^a *Department of Biostatistics, University at Buffalo, The State University of New York, United States*

^b *Department of Statistics, University of Rochester, United States*

Received 9 October 2006; received in revised form 10 July 2007; accepted 31 July 2007

Abstract

The sampling distributions are generally unavailable in exact form and are approximated either in terms of the asymptotic distributions, or their correction using expansions such as Edgeworth, Laguerre or Cornish–Fisher; or by using transformations analogous to that of Wilson and Hilferty. However, when theoretical routes are intractable, in this electronic age, the sampling distributions can be reasonably approximated using empirical methods. The point is illustrated using the null distribution of Hoeffding's test of bivariate independence which is important because of its consistency against all dependence alternatives. For constructing the approximations we employ two Weibull extensions, the generalized Weibull and the exponentiated Weibull families, which contain a rich variety of density shapes and tail lengths, and have their distribution functions and quantile functions available in closed form, making them convenient for obtaining the necessary percentiles and p -values. Both approximations are seen to be excellent in terms of accuracy, but that based on the generalized Weibull is more portable.

© 2007 Elsevier B.V. All rights reserved.

Keywords: Hoeffding's test of independence; Weibull extensions

1. Introduction

In the vast array of problems encountered in statistical analysis the necessary sampling distributions are rarely available in exact form. They are often replaced by their asymptotic versions, which are in general unsatisfactory when the samples are small or of moderate size, or the asymptotics are refined in order to obtain reasonable approximations. The percentiles

* Corresponding author. Tel.: +1 716 829 2753; fax: +1 716 829 2200.

E-mail address: gwilding@buffalo.edu (G.E. Wilding).

of the asymptotic distribution are corrected using Cornish–Fisher type series, and probabilities using Edgeworth or Laguerre type expansions. The convergence to asymptotic distributions is also accelerated using transformations such as Fisher’s Z of the correlation coefficient, and the analogues of Wilson and Hilferty’s cube root developed by Mudholkar and Trivedi [17] for the sample variance, and by Mudholkar, Trivedi and van der Meulen [18] for a variety of likelihood ratios. When the theory routes are considered intractable, Monte Carlo percentiles are used, which are not entirely satisfactory for obtaining p -values.

The purpose of this paper is to describe how, in this electronic age, largely empirical methods can be used to construct good approximations for the sampling distributions. Towards this end the null distribution of Hoeffding’s [5] classical rank test of bivariate independence, known to be consistent against all dependence alternatives, is approximated by members of two Weibull extensions known as the generalized Weibull and the exponentiated Weibull families. Hoeffding’s rank test, for which the null distribution is unavailable, together with its usage is outlined in Section 2. The two extended Weibull families, which contain distributions with a rich variety of density shapes and tail lengths, have their distribution functions and quantile functions available in closed forms making them suitable for estimating both percentiles and p -values. They are described in Section 3. The development of the approximations and evaluation are discussed in Section 4. An example is discussed in Section 5 and the final section contains conclusions.

2. The consistent tests of independence

Although the assumption of bivariate independence is commonly assessed using the Pearson correlation coefficient, it is generally recognized that it may not be the most appropriate test, as it lacks consistency against all dependence alternatives. Hence a variety of other measures of dependence and corresponding tests have appeared in the literature which is substantial. We note that most of these tests are also not consistent against all dependence alternatives.

Hoeffding’s [5] nonparametric rank test for bivariate independence, which is consistent against all bivariate dependence alternatives, is based on the simple notion that

$$D(x, y) = F_{X,Y}(x, y) - F_X(x)F_Y(y) = 0, \quad (2.1)$$

if and only if X and Y are independent. Nonparametric estimation of the quantity $\int D^2(x, y)dF(x, y)$ results in the statistic,

$$D_n = \frac{Q - 2(n-2)R + (n-2)(n-3)S}{n(n-1)(n-2)(n-3)(n-4)}, \quad (2.2)$$

where

$$Q = \sum_{i=1}^n (R_i - 1)(R_i - 2)(S_i - 1)(S_i - 2), \quad (2.3)$$

$$R = \sum_{i=1}^n (R_i - 2)(S_i - 2)c_i, \quad (2.4)$$

$$S = \sum_{i=1}^n (c_i - 1)c_i, \quad (2.5)$$

R_i and S_i being the respective ranks of X_i among the X ’s and Y_i among the Y ’s, and c_i is the number of bivariate observations (X_j, Y_j) for which $X_j \leq X_i$ and $Y_j \leq Y_i$. The statistic D_n is

bounded by the interval $[-\frac{1}{60}, \frac{1}{30}]$, where the upper bound is fixed for all n , and the lower bound of nD_n increases to $-1/36$ as $n \rightarrow \infty$.

Hoeffding obtained the exact null distribution of D_n for $n = 5, 6, 7$. He also obtained the characteristic function of the asymptotic null distribution of $nD_n + (1/36)$ but found it computationally inconvenient. This motivated Blum, Kiefer and Rosenblatt [1] to consider the simpler statistic,

$$B_n = n^{-5} \sum_{i=1}^n [N_1(i)N_4(i) - N_2(i)N_3(i)]^2, \quad (2.6)$$

where $N_1(i)$, $N_2(i)$, $N_3(i)$, and $N_4(i)$ are the number of points lying in the four quadrants determined by the vertical and horizontal lines through the point (X_i, Y_i) . The statistic nB_n is asymptotically equivalent to $nD_n + (1/36)$. Blum, Kiefer and Rosenblatt obtained the Laplace transform of the asymptotic null distribution of nB_n , and gave methods for approximating the asymptotic null distribution. They also gave a tabulation of the approximate percentiles usable in large samples.

The belief that the asymptotic distribution of nD_n is adequately close to finite sample versions has led to its use for testing bivariate independence in moderate size, and even small size, samples. Such use of the asymptotic distribution appears in commonly used programs such as the SAS procedure PROC CORR, see SAS Procedures Guide [21], to obtain p -values for Hoeffding's D_n statistic.

Though the tests of bivariate independence based on nD_n and nB_n have been largely ignored in applications, some useful discussion may be found in Hollander and Wolfe [6]. Further examination of the two consistent tests, including finite sample percentile tabulation and a Monte Carlo study of the power functions based on random generation from a variety of bivariate copula families, appears in Mudholkar and Wilding [19]. Applications of both the consistent tests in connection with goodness-of-fit problems may be found in Wilding and Mudholkar [20] and Marchetti, Mudholkar and Wilding [9]. Though the implementation of the two consistent tests of independence is simplified via tables of percentiles given in Mudholkar and Wilding [19], lack of distributional approximations makes it difficult to obtain p -values except by rough interpolation.

3. The two Weibull extensions

The two extensions of the Weibull family which are used for approximating the null distribution of Hoeffding's test of independence are now described.

The generalized Weibull family. The generalized Weibull family, first suggested in Mudholkar et al. [12] for constructing isotones, has its quantile function $Q(u)$ ($0 \leq u \leq 1$), distribution function $F(x)$, and its density function $f(x)$ given respectively by,

$$Q(u) = \begin{cases} \sigma[(1 - (1 - u)^\lambda)/\lambda]^\gamma & \lambda \neq 0, \\ \sigma[-\log(1 - u)]^\gamma & \lambda = 0, \end{cases} \quad (3.1)$$

$$F(x) = \begin{cases} 1 - (1 - \lambda(x/\sigma)^{1/\gamma})^{1/\lambda} & \lambda \neq 0, \\ 1 - \exp(-(x/\sigma)^{1/\gamma}) & \lambda = 0, \end{cases} \quad (3.2)$$

and

$$f(x) = \begin{cases} \frac{1}{\gamma\sigma} [1 - \lambda(x/\sigma)^{1/\gamma}]^{1/\lambda-1} (x/\sigma)^{1/\gamma-1} & \lambda \neq 0, \\ \frac{1}{\gamma\sigma} (x/\sigma)^{(1/\gamma-1)} \exp(-(x/\sigma)^{1/\gamma}) & \lambda = 0, \end{cases} \quad (3.3)$$

where $\gamma, \sigma > 0$ and $-\infty < \lambda < \infty$. The range of the generalized Weibull random variable x is $(0, \infty)$ in the case of $\lambda \leq 0$ and in the case of $\lambda > 0$ the range is $(0, \sigma/\lambda^\gamma)$. The generalized Weibull family yields the Weibull family when $\lambda = 0$, the exponential distribution for $\gamma = 1$, $\lambda = 0$, and the log-logistic distribution for $\lambda = -1$, which is often used as a model in survival studies. Moreover, common parametric alternatives to exponentiality, e.g. the lognormal and the gamma distributions, are very well-approximated by members of the family, see Mudholkar and Kollia [11]. Thus the distribution corresponding to $(\gamma, \lambda) = (1.424, 0.057)$ agrees in density shape and in skewness and kurtosis with the χ_1^2 distribution. Similarly, the distribution with shape parameters $(\gamma, \lambda) = (0.82, -0.033)$ represents χ_3^2 , with $(\gamma, \lambda) = (0.60, -0.084)$ it is like χ_6^2 , and with $(\gamma, \lambda) = (0.534, -0.11)$ it is like the χ_8^2 distribution.

Further analysis of the generalized Weibull family, including examination of the skewness and kurtosis, density shapes and tail characteristics, extreme value distributions, density classification, and its relation to the Pearson system and other distributions can be found in Mudholkar and Kollia [11]. Applications may also be found in Mudholkar, Srivastava and Kollia [16] and Mudholkar and Sarkar [13].

The exponentiated Weibull family. Originally proposed by Mudholkar and Srivastava [14] in the context of application to bathtub shaped failure rate data, the exponentiated Weibull has quantile function,

$$Q(u) = \sigma[-\log(1 - u^{1/\theta})]^{1/\alpha}, \quad (3.4)$$

distribution function,

$$F(x) = [1 - \exp(-(x/\sigma)^\alpha)]^\theta, \quad (3.5)$$

and density function,

$$f(x) = \frac{\alpha\theta}{\sigma} [1 - \exp(-(x/\sigma)^\alpha)]^{\theta-1} \exp(-(x/\sigma)^\alpha) (x/\sigma)^{\alpha-1}, \quad (3.6)$$

where $\alpha, \theta, \sigma > 0$, $0 \leq u \leq 1$, and $0 \leq x < \infty$. Distributional properties, including skewness and kurtosis, density shapes and tail character, and the extreme value and extreme spacings distributions in the exponentiated Weibull family are found in Mudholkar and Hutson [10], which also includes an application to analysis of flood data, demonstrating the usefulness of the family in connection with the extreme value data. Other references include Mudholkar, Srivastava and Freimer [15], Gupta et al. [3], Gupta and Kundu [4], and Jiang and Murthy [7].

In modeling and analysis of life-time data appearing in survival studies and reliability analysis, the hazard rate, i.e. failure rate, is a critical consideration. In terms of the hazard rate characteristics, the generalized Weibull and exponentiated Weibull families are rich, containing not only increasing and decreasing failure rate distributions, but also distributions with unimodal and bathtub shaped hazard functions. It is seen that the hazard functions in both families assume these four hazard shapes in four well-delineated regions surrounding a point representing the exponential distribution in the planes of the shape parameters of these families. Additionally, both

families have quantile functions which are in closed form. Therefore these two families have been found to be ideal for use in construction of isotones for goodness-of-fit tests of exponentiality, see Wilding, Mudholkar, and Kollia [22], and now also for the development of distributional approximations.

4. Approximations for the null distribution

Use of the two Weibull extensions for approximating sampling distributions is now illustrated.

4.1. Asymptotic case

Since $nD_n + (1/36)$ is asymptotically equivalent to nB_n , approximate asymptotic percentiles of nD_n may be computed from the percentiles given in Blum, Kiefer and Rosenblatt [1]. Modeling the null distribution using these percentiles and corresponding probabilities in terms of the generalized Weibull quantile function shifted by a location parameter, that is,

$$Q(u) = \mu + \sigma[(1 - (1 - u)^\lambda)/\lambda]^\gamma, \quad (4.1)$$

can be accomplished using S-plus nonlinear least-squares routine *nlregb*. The parameters of the fit of (4.1) so obtained are:

$$\sigma_g = 0.1435, \quad \lambda_g = -7.26, \quad \gamma_g = 0.01266, \quad \mu_g = -0.1537, \quad (4.2)$$

where the subscript “g” indicates the generalized Weibull family. Similar nonlinear regression with the exponentiated Weibull quantile function with an additional location parameter, that is,

$$Q(u) = \mu + \sigma[-\log(1 - u^{1/\theta})]^{1/\alpha}, \quad (4.3)$$

results in the fit given by parameter values,

$$\sigma_e = 6.39\text{E-}4, \quad \theta_e = 51.7, \quad \alpha_e = 0.4364, \quad \mu_e = -0.0226, \quad (4.4)$$

where the subscript “e” signifies the exponentiated Weibull family. The asymptotic null distribution of nD_n is excellently approximated by the distributions with quantile functions (4.1) and (4.3), where the parameters are given respectively by (4.2) and (4.4). Fig. 1 displays the quality of these fits to the percentiles given by Blum, Kiefer and Rosenblatt [1]. Note that the two quantile functions are indistinguishable in the figure.

4.2. Finite samples

In order to obtain the approximation based on the Weibull extensions for moderate size samples, empirical estimates of the percentiles corresponding to those in Blum, Kiefer and Rosenblatt, for $n = 5(1)25(5)50(10)100, 150, 200(100)500, 750, 1000$ were obtained using Monte Carlo simulations with 10,000 replications each. For each n , parameter estimates for the models (4.1) and (4.3) were obtained using S-plus nonlinear regression routine *nlregb* as in the asymptotic case discussed above. Both Weibull extensions are known to contain a wide variety of distributional shapes, therefore a given set of percentiles gives reasonable fits in more than one region in the parametric space. Hence, the choice of starting value for the iterative procedure is important. Also, due to the discreteness of the statistic nD_n , parameter values may sometimes “jump around” as n increases. It is also noted that, since the Weibull extensions are only defined

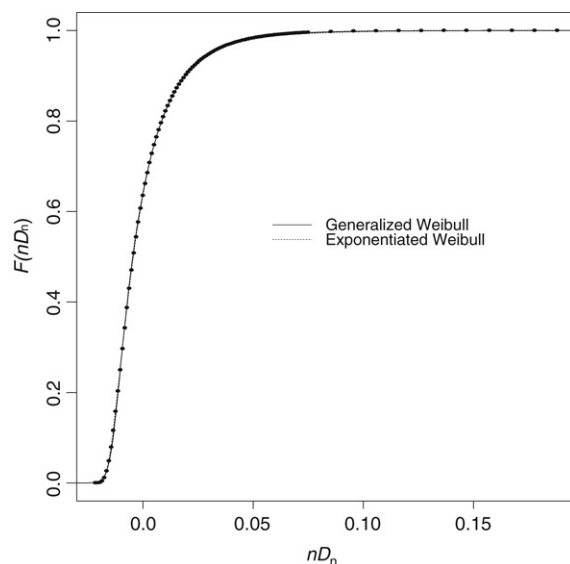


Fig. 1. Generalized Weibull and exponentiated Weibull fit of asymptotic percentiles¹ of nD_n .

over a positive range, the upper bound used in the nonlinear regression for the shift parameter was empirically estimated for each n . The parameter estimates of the fits for both families are presented in Table 1.

4.3. Validation

Table 2 presents a comparison of the approximate probabilities of nD_n obtained using the approximations in terms of the models at (4.1) and (4.3), using the parameter values given by (4.2), (4.4) and Table 1. Also included are probability approximations obtained, as in SAS procedure PROC CORR (thanks to their technical support staff), by interpolating percentiles given in [1]. The nominal percentiles for the finite sample comparisons were obtained empirically from another Monte Carlo experiment based on 100,000 replications. It can be seen that both approximations based on the Weibull extensions are of comparable accuracy over a wide range of the sample sizes and probabilities. The accuracy in the right tail of the null distribution, which is more relevant to statistical decisions since the test statistic rejects for large values, is of much higher quality than that in the left tail. It can also be seen that the interpolation approach taken by SAS does not fair well in the right tail for small and moderate sized samples and does extremely poor in the left tail, even for larger sample sizes. That is, using the asymptotic percentiles in the approximation of the corresponding finite sample quantities performs poorly.

4.4. Portable modification

In order to make the approximations described in Section 4.2 more portable, nonlinear least-squares fits for the parameters presented in Table 1 as functions of n were attempted. In the

¹ Blum, Kiefer and Rosenblatt [1] tabulation.

Table 1
Parameters of the generalized Weibull and exponentiated Weibull approximations to the null distribution of nD_n

n	Generalized Weibull				Exponentiated Weibull			
	$\sigma_{g,n}$	$\lambda_{g,n}$	$\gamma_{g,n}$	$\mu_{g,n}$	$\sigma_{e,n}$	$\theta_{e,n}$	$\alpha_{e,n}$	$\mu_{e,n}$
10	0.2649	−3.5764	0.02352	−0.2785	1.43E−04	77,969.1	0.3852	−0.0902
11	0.1438	−3.2415	0.04336	−0.1559	1.09E−04	77,969.1	0.3738	−0.0830
12	0.2352	−3.6884	0.02313	−0.2477	1.21E−04	77,961.9	0.3822	−0.0803
13	0.2347	−4.1184	0.02085	−0.2476	4.42E−05	77,961.8	0.3373	−0.0696
14	0.2403	−4.2806	0.01917	−0.2528	2.97E−05	77,957.7	0.3231	−0.0646
15	0.1777	−4.0286	0.02608	−0.1898	2.49E−05	77,957.7	0.3177	−0.0620
16	0.1649	−3.9619	0.02762	−0.1765	2.88E−05	77,954.2	0.3241	−0.0614
17	0.1727	−4.1650	0.02430	−0.1844	2.85E−05	77,954.2	0.3255	−0.0591
18	0.1169	−4.0818	0.03592	−0.1283	8.02E−06	77,950.6	0.2831	−0.0520
19	0.1270	−3.6513	0.03513	−0.1375	4.53E−05	53,499.1	0.3422	−0.0587
20	0.1570	−4.4487	0.02454	−0.1685	1.15E−05	53,499.1	0.2925	−0.0503
21	0.2276	−5.0676	0.01533	−0.2396	1.41E−05	39,039.0	0.2974	−0.0491
22	0.1279	−3.9699	0.03124	−0.1384	3.58E−05	39,038.3	0.3322	−0.0529
23	0.2049	−4.9117	0.01687	−0.2162	1.69E−05	39,038.7	0.3046	−0.0485
24	0.2246	−5.1171	0.01471	−0.2362	1.63E−05	39,036.0	0.3039	−0.0478
25	0.1394	−4.8334	0.02412	−0.1507	2.82E−05	7522.9	0.3106	−0.0421
30	0.1595	−4.9067	0.01984	−0.1705	1.94E−04	1377.5	0.3886	−0.0407
35	0.1620	−5.1764	0.01834	−0.1729	1.11E−04	1627.1	0.3597	−0.0379
40	0.1062	−4.6415	0.02942	−0.1162	3.35E−05	5661.7	0.3185	−0.0380
45	0.2264	−6.3728	0.01068	−0.2378	6.95E−05	1524.1	0.3380	−0.0340
50	0.1719	−5.9549	0.01454	−0.1829	2.78E−04	303.3	0.3952	−0.0316
60	0.1392	−5.3732	0.01856	−0.1495	9.69E−05	1177.3	0.3541	−0.0328
70	0.1664	−5.9154	0.01471	−0.1767	4.40E−04	180.1	0.4225	−0.0299
80	0.2461	−6.6774	0.00898	−0.2570	9.15E−05	852.7	0.3469	−0.0308
90	0.1507	−5.8640	0.01598	−0.1608	4.36E−04	155.8	0.4202	−0.0285
100	0.1469	−5.8101	0.01599	−0.1568	1.15E−03	47.2	0.4908	−0.0260
150	0.3388	−7.4224	0.00564	−0.3497	5.49E−04	94.6	0.4347	−0.0261
200	0.1227	−6.6537	0.01626	−0.1331	8.21E−05	523.1	0.3383	−0.0267
300	0.1488	−6.8793	0.01311	−0.1590	3.78E−04	109.9	0.4072	−0.0247
400	0.1477	−6.8496	0.01338	−0.1578	6.59E−04	59.5	0.4401	−0.0239
500	0.2010	−7.5387	0.00908	−0.2114	2.14E−04	191.3	0.3773	−0.0251
750	0.1682	−6.8899	0.01143	−0.1783	1.59E−03	22.1	0.5154	−0.0221
1000	0.1761	−7.6529	0.01004	−0.1866	9.77E−04	32.8	0.4662	−0.0222

regression of each column in Table 1 on n , nonlinear equations used were chosen such that the limiting values as $n \rightarrow \infty$ were the parameter values obtained in Section 4.1 based on the asymptotic percentiles. In practice, the resulting fits may be used in place of the values contained within Table 1. The attempt was reasonably successful in the case of the shifted generalized Weibull approximation, leading to equations,

$$\sigma_{g,n} = \sigma_g + 0.0385 \exp(-0.0019n) + 0.38 \exp(-0.174n), \tag{4.5}$$

$$\lambda_{g,n} = \lambda_g / (1 + 2.09 \exp(-0.1n) - 0.3 \exp(-0.0025n)), \tag{4.6}$$

$$\gamma_{g,n} = \gamma_g + 0.0037 \exp(-0.0114n) + 0.023 \exp(-0.088n), \tag{4.7}$$

$$\mu_{g,n} = \mu_g (1 + 0.253 \exp(-0.002n) + 2.43 \exp(-0.168n)). \tag{4.8}$$

However, the effort in the case of the exponentiated Weibull approximation, subject to achieving reasonable accuracy, gave relatively intricate and nonelegant equations and was not pursued. In summary, the following quantile, distribution and density functions provide a reasonable

Table 2

A comparison of the nominal and approximate probabilities of nD_n as approximated by the generalized Weibull (GW), the exponentiated Weibull (EW) and by linear interpolation of asymptotic percentiles (SAS)

n		Nominal probabilities										
		0.010	0.025	0.050	0.100	0.250	0.500	0.750	0.900	0.950	0.975	0.990
Approximations												
10	GW	0.015	0.029	0.044	0.094	0.263	0.498	0.752	0.898	0.951	0.976	0.990
	EW	0.007	0.024	0.045	0.109	0.284	0.498	0.746	0.899	0.953	0.977	0.991
	SAS	—	—	—	—	0.048	0.441	0.814	0.948	0.981	0.993	0.998
20	GW	0.015	0.026	0.046	0.086	0.237	0.505	0.751	0.898	0.948	0.973	0.988
	EW	0.008	0.022	0.049	0.099	0.255	0.502	0.747	0.898	0.948	0.973	0.988
	SAS	—	—	0.000	0.003	0.124	0.480	0.783	0.928	0.968	0.986	0.995
50	GW	0.014	0.025	0.046	0.089	0.247	0.511	0.753	0.901	0.951	0.975	0.990
	EW	0.009	0.025	0.052	0.103	0.255	0.503	0.751	0.901	0.950	0.974	0.988
	SAS	0.000	0.002	0.010	0.042	0.198	0.492	0.764	0.912	0.959	0.980	0.993
100	GW	0.014	0.026	0.043	0.083	0.234	0.502	0.753	0.902	0.952	0.976	0.990
	EW	0.009	0.024	0.048	0.097	0.246	0.494	0.750	0.903	0.953	0.976	0.990
	SAS	0.002	0.009	0.024	0.067	0.224	0.497	0.758	0.906	0.955	0.978	0.991
150	GW	0.013	0.025	0.045	0.093	0.255	0.512	0.753	0.905	0.956	0.980	0.993
	EW	0.010	0.025	0.052	0.105	0.255	0.504	0.755	0.905	0.953	0.976	0.989
	SAS	0.003	0.011	0.030	0.077	0.231	0.497	0.754	0.904	0.954	0.977	0.991
200	GW	0.014	0.026	0.045	0.090	0.250	0.514	0.755	0.901	0.950	0.974	0.989
	EW	0.010	0.026	0.051	0.103	0.254	0.506	0.755	0.901	0.949	0.972	0.986
	SAS	0.005	0.016	0.036	0.083	0.235	0.497	0.752	0.902	0.952	0.976	0.991
500	GW	0.014	0.024	0.044	0.087	0.249	0.504	0.747	0.899	0.951	0.976	0.991
	EW	0.010	0.025	0.050	0.098	0.250	0.496	0.749	0.899	0.948	0.971	0.987
	SAS	0.008	0.021	0.044	0.092	0.247	0.500	0.752	0.901	0.951	0.975	0.991
∞	GW	0.014	0.025	0.045	0.090	0.247	0.506	0.749	0.900	0.950	0.975	0.990
	EW	0.009	0.023	0.050	0.102	0.251	0.498	0.749	0.900	0.949	0.973	0.987

“–”: Statistic outside the range of asymptotic distribution.

approximation to the null distribution of nD_n :

$$Q_n(u) = \mu_{g,n} + \sigma_{g,n} \left(\frac{1 - (1 - u)^{\lambda_{g,n}}}{\lambda_{g,n}} \right)^{\gamma_{g,n}}, \quad (4.9)$$

$$F_n(x) = 1 - \left(1 - \lambda_{g,n} \left(\frac{x - \mu_{g,n}}{\sigma_{g,n}} \right)^{1/\gamma_{g,n}} \right)^{1/\lambda_{g,n}}, \quad (4.10)$$

$$f_n(x) = \frac{1}{\gamma_{g,n} \sigma_{g,n}} \left(1 - \lambda_{g,n} \left(\frac{x - \mu_{g,n}}{\sigma_{g,n}} \right)^{1/\gamma_{g,n}} \right)^{1/\lambda_{g,n} - 1} \left(\frac{x - \mu_{g,n}}{\sigma_{g,n}} \right)^{1/\gamma_{g,n} - 1}, \quad (4.11)$$

where the parameter values are given by (4.5)–(4.8).

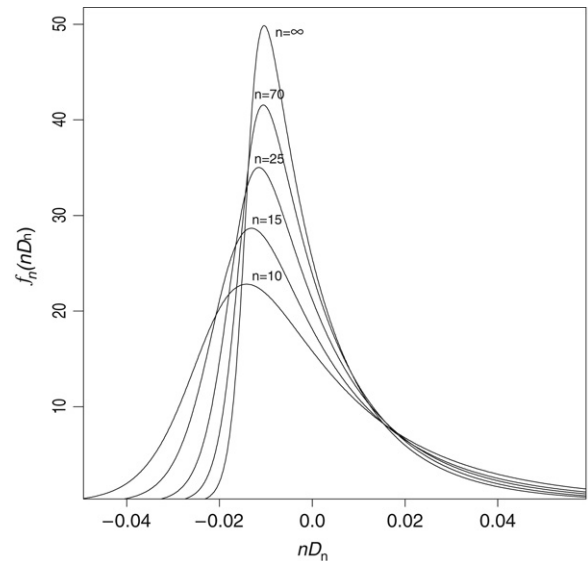


Fig. 2. Approximate² probability densities of nD_n .

Table 3
A comparison of the nominal and approximate probabilities of nD_n as approximated by the generalized Weibull distribution; estimated parameters

n	Nominal probabilities										
	0.010	0.025	0.050	0.100	0.250	0.500	0.750	0.900	0.950	0.975	0.990
Approximations											
10	0.015	0.029	0.045	0.095	0.264	0.497	0.751	0.897	0.950	0.975	0.990
20	0.015	0.027	0.047	0.089	0.242	0.508	0.752	0.899	0.950	0.974	0.989
50	0.015	0.026	0.045	0.085	0.232	0.492	0.740	0.894	0.947	0.973	0.989
100	0.018	0.032	0.052	0.094	0.245	0.502	0.747	0.897	0.949	0.974	0.990
150	0.017	0.030	0.050	0.094	0.243	0.501	0.747	0.899	0.951	0.976	0.990

The quantile function and the distribution function can be used to obtain the percentiles and p -values respectively. The density function approximation (4.11) is used to construct Fig. 2 which provides a view of the p.d.f.'s of the null distribution of nD_n .

Evaluation of portable modification. To assess the adequacy of the generalized Weibull approximation as given by nonlinear equations (4.5)–(4.8), a comparison of the nominal and approximate probabilities is given in Table 3. Although it is expected that some loss of accuracy would occur when estimating the parameters as a function of n as compared to use of the raw parameter estimates, evaluation of the approximation reveals adequate fit for samples sizes 10 to ∞ , and over the full range of the nD_n statistic.

5. An example

We now consider an example discussed within Lehmann [8, Chapter 7, Problem 19] based on the work of Clark, Vandenberg, and Proctor [2]. The data consists of pairs of psychological

² Generalized Weibull approximation based on (4.5)–(4.8).

Table 4
Psychological test scores from 21 identical twins

45	40	48	42	45	44	40	37	27	45	51	44	44	29	27	48	33	39	47	54	32
37	41	50	46	31	39	37	42	27	48	49	52	27	36	29	44	44	43	34	48	33

test scores obtained from a total of 21 identical twin males. The raw data appears in Table 4. After running SAS PROC CORR with the HOEFFDING option, the p -value corresponding to Hoeffding's test of independence was seen to be 0.049. We note that the value of the test statistic presented in the output, 0.046, is D_n after having been multiplied by 30 so that the range of the statistic is no more than 1 and no less than -0.5 . After applying the approximation presented in this paper, we find the p -value to be 0.179. We see the decision to reject or not is inconsistent between the two methods when using a nominal significance level of 0.05. This demonstrates the pitfalls of using the asymptotic percentiles when performing Hoeffding's test and makes a case for using the results presented in this paper.

6. Conclusions

In this paper we have discussed how empirical methods can be used to construct distributional approximations, especially when theoretical routes appear blocked. In the process convenient and accurate approximations for Hoeffding's statistic nD_n for testing bivariate independence are developed.

Acknowledgment

The authors would like to thank Mike Stockstill of SAS technical support for his assistance in obtaining the probability approximations in Table 2.

References

- [1] J.R. Blum, J. Kiefer, M. Rosenblatt, Distribution free tests of independence based on the sample distribution function, *The Annals of Mathematical Statistics* 32 (1961) 485–498.
- [2] P. Clark, S. Vandenberg, C. Proctor, On the relationship of scores on certain psychological tests with a number of anthropometric characteristics and birth order in twins, *Human Biology* 33 (1961) 163–180.
- [3] R.C. Gupta, P.L. Gupta, R.D. Gupta, Modeling failure time data by Lehman alternatives, *Communications in Statistics. Theory and Methods* 27 (1998) 887–904.
- [4] R.D. Gupta, D. Kundu, Generalized exponential distributions, *Australian and New Zealand Journal of Statistics* 41 (1999) 173–188.
- [5] W. Hoeffding, A nonparametric test of independence, *The Annals of Mathematical Statistics* 19 (1948) 546–557.
- [6] M. Hollander, D.A. Wolfe, *Nonparametric Statistical Methods*, Wiley, New York, 1999.
- [7] R. Jiang, D.N.P. Murthy, The exponentiated Weibull family: A graphical approach, *IEEE Transactions on Reliability* 48 (1999) 68–72.
- [8] E.L. Lehmann, *Nonparametrics: Statistical Methods Based on Ranks*, Prentice-Hall, New Jersey, 1998.
- [9] C.E. Marchetti, G.S. Mudholkar, G.E. Wilding, Testing goodness-of-fit of the gamma models, in: Y.P. Chaubey (Ed.), *Recent Advances in Statistical Methods*, World Scientific Publishing Company, Inc., London, 2002.
- [10] G.S. Mudholkar, A.D. Hutson, The exponentiated Weibull family: Some properties and a flood data application, *Communications in Statistics. Theory and Methods* 25 (1996) 3059–3083.
- [11] G.S. Mudholkar, G.D. Kollia, Generalized Weibull family: A structural analysis, *Communications in Statistics. Theory and Methods* 23 (1994) 1149–1171.
- [12] G.S. Mudholkar, G.D. Kollia, C.T. Lin, K.R. Patel, A graphical procedure for comparing goodness-of-fit tests, *Journal of the Royal Statistical Society B* 53 (1991) 221–232.

- [13] G.S. Mudholkar, I.C. Sarkar, A proportional hazards modeling of multisample reliability data, *Communications in Statistics. Theory and Methods* 28 (1999) 2079–2101.
- [14] G.S. Mudholkar, D.K. Srivastava, Exponentiated Weibull family for analyzing bathtub failure-rate data, *IEEE Transactions on Reliability* 42 (1993) 299–302.
- [15] G.S. Mudholkar, D.K. Srivastava, M. Freimer, The exponentiated Weibull family: A reanalysis of the bus-motor-failure data, *Technometrics* 37 (1995) 436–445.
- [16] G.S. Mudholkar, D.K. Srivastava, G.D. Kollia, A generalization of the Weibull distribution with applications to the analysis of survival data, *Journal of the American Statistical Association* 91 (1996) 1575–1583.
- [17] G.S. Mudholkar, M.C. Trivedi, A Gaussian approximation to the distribution of the sample variance for nonnormal populations, *Journal of the American Statistical Association* 76 (1981) 479–485.
- [18] G.S. Mudholkar, M.C. Trivedi, E.C. van der Meulen, A gamma analogue of the Wilson–Hilferty transformation, *Communications in Statistics. Simulation and Computation* 27 (1998) 21–37.
- [19] G.S. Mudholkar, G.E. Wilding, On the conventional wisdom regarding two consistent tests of bivariate independence, *Journal of the Royal Statistical Society, Series D* 52 (2003) 41–57.
- [20] G.E. Wilding, G.S. Mudholkar, Some modifications of the Z-tests of normality and their isotones, *Statistical Methodology* (in press).
- [21] SAS, SAS Procedures Guide, Version 6, third edition, SAS Institute Inc., Cary, NC, 1990.
- [22] G.E. Wilding, G.S. Mudholkar, G.D. Kollia, Two sets of isotones for comparing tests of exponentiality, *Journal of Statistical Planning and Inference* 137 (2007) 3815–3825.